

pqcoin: A Post-Quantum Cryptocurrency

Version 0.1.0

Abstract

pqcoin is a proof-of-work cryptocurrency designed to provide security against both classical and quantum computers. By utilizing NIST-standardized post-quantum cryptographic primitives - specifically ML-DSA-87 (FIPS 204) for digital signatures and SHA3-512 (FIPS 202) for hashing - pqcoin offers a quantum-resistant alternative to existing cryptocurrencies that rely on elliptic curve cryptography.

The design follows Bitcoin's proven UTXO model while making deliberate simplifications: removing the scripting language in favor of two well-defined output types (P2PKH and M-of-N multisig), and modernizing the cryptographic foundation for the post-quantum era.

1. Introduction

1.1 Motivation

Current cryptocurrencies like Bitcoin rely on the Elliptic Curve Digital Signature Algorithm (ECDSA) for transaction authorization. While secure against classical computers, ECDSA is vulnerable to Shor's algorithm running on a sufficiently powerful quantum computer. As quantum computing advances, the cryptocurrency ecosystem faces an existential threat to its security model.

pqcoin addresses this threat by building on cryptographic primitives that are believed to be secure against both classical and quantum attacks, as standardized by NIST in their Post-Quantum Cryptography project.

1.2 Design Goals

1. Quantum Resistance: Use NIST-standardized post-quantum cryptographic algorithms
2. Simplicity: Remove unnecessary complexity while maintaining Bitcoin's proven security model
3. Correctness: Prioritize security and correctness over feature richness
4. Compatibility: Follow familiar patterns from Bitcoin where appropriate

2. Cryptographic Primitives

2.1 Hash Function: SHA3-512 (FIPS 202)

All hashing operations in pqcoin use SHA3-512, providing 512-bit output (64 bytes), 256-bit classical security, and ~256-bit security against Grover's algorithm.

SHA3-512 is used for: Transaction ID computation, Block header hashing (proof-of-work), Address derivation (hash of public key), and Merkle tree construction.

2.2 Digital Signatures: ML-DSA-87 (FIPS 204)

Transaction authorization uses ML-DSA-87 (formerly Dilithium5), a lattice-based digital signature scheme providing NIST Level 5 security (~256-bit classical, ~128-bit quantum security).

Key sizes: Public Key = 2,592 bytes, Secret Key = 4,896 bytes, Signature = 4,627 bytes

3. Transaction Model

3.1 UTXO Model

pqcoin uses the Unspent Transaction Output (UTXO) model: Transactions consume existing outputs (inputs) and create new outputs. Each output can only be spent once. The sum of input values must equal or exceed output values. The difference is the transaction fee.

3.2 Output Types

Unlike Bitcoin's scripting system, pqcoin supports exactly two output types: Pay-to-Public-Key-Hash (P2PKH) for single-signature outputs, and M-of-N Multisig for multi-signature outputs (limited to 15 keys due to post-quantum signature sizes).

4. Block Structure

The block header is 176 bytes: version (4 bytes), prev_hash (64 bytes), merkle_root (64 bytes), timestamp (8 bytes), difficulty_bits (4 bytes), nonce (32 bytes).

pqcoin uses a 256-bit nonce, providing 2^{256} possible values. This eliminates nonce exhaustion concerns and removes the need for Bitcoin-style extraNonce mechanisms.

Block limits: Max 1,000 transactions, 16 MB max block size, 10,000 max inputs/outputs per transaction, 15 max multisig keys.

5. Consensus Mechanism

Valid blocks must have $\text{SHA3-512}(\text{block_header}) \leq \text{target}$. Difficulty adjusts every 2,016 blocks to maintain 10-minute average block time, with adjustment clamped to 4x in either direction. The chain with most cumulative proof-of-work is canonical. Reorganizations are limited to 100 blocks.

6. Monetary Policy

Initial reward: 50,000,000 quanta (50 coins). Halving interval: 210,000 blocks. Total supply asymptotically approaches 21 million coins. Coinbase maturity: 100 blocks.

7. Network Protocol

Messages use magic bytes 'PQCN', with length-prefixed framing and SHA3-512 checksum. Message types include version/verack, ping/pong, inv/getdata, getheaders/headers, block/tx, and addr/addrv2.

Network parameters: Default port 8333, max 125 connections (8 outbound), max 3 per IP, max 2 per /16 subnet, 24-hour ban duration.

8. Security Considerations

Quantum resistance: ML-DSA-87 (based on Module-LWE problem) and SHA3-512 (Grover's algorithm provides only quadratic speedup). Classical security: 256-bit hash and signature security.

Known limitations: ML-DSA-87 signatures are ~72x larger than ECDSA (4,627 vs 64 bytes), no scripting support, and mempool does not support chained unconfirmed transactions.

9. Comparison with Bitcoin

pqcoin vs Bitcoin:

- Hash: SHA3-512 (64 bytes) vs SHA-256 (32 bytes)
- Signatures: ML-DSA-87 vs ECDSA
- Signature size: 4,627 bytes vs 64 bytes
- Quantum resistant: Yes vs No
- Scripting: No (P2PKH + Multisig only) vs Yes
- Nonce: 256 bits vs 32 bits
- Block size: 16 MB vs ~4 MB

10. Conclusion

pqcoin provides a quantum-resistant cryptocurrency built on NIST-standardized cryptographic primitives. By simplifying Bitcoin's design while upgrading its cryptographic foundation, pqcoin offers a secure platform for value transfer in the post-quantum era.

References

1. NIST FIPS 202: SHA-3 Standard
2. NIST FIPS 204: Module-Lattice-Based Digital Signature Standard

3. Nakamoto, S. (2008). Bitcoin: A Peer-to-Peer Electronic Cash System
4. NIST Post-Quantum Cryptography Standardization Process